

The University of the South Pacific

Serving the Cook Islands, Fiji, Kiribati, Marshall Islands, Nauru, Niue, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu, and Vanuatu.

School of Information Technology, Engineering, Mathematics and Physics

CS324: Distributed System

Test 1 – Semester 2, 2025

Learning Outcome: Outline the importance of distributed systems over stand-alone systems

Time Allowed: 50 mins

Total Marks: 30 (10% of final grade)

Name: _____

Student ID: _____

INSTRUCTIONS:

- All questions are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

Question 1: Explain with reasons whether the following scenarios refer to distributed, centralized, or decentralized systems. (4 marks)

Scenario 1: A university library maintains one central server where all students access e-books. If the server goes down, no student can access resources.

Scenario 2: A ride-sharing application where drivers and passengers are matched using a distributed set of servers across different regions to balance the load.

ANS: Answer: Centralized system – because all services depend on one central server. Failure of the server leads to total unavailability.

Answer: Distributed system – because multiple servers coordinate and share load to ensure reliability and scalability.

Question 2: An IoT-based **smart city traffic management system** collects data from thousands of sensors placed on roads, traffic lights, and vehicles. The system must process this information in real time to:

- detect congestion,
- notify drivers about alternative routes, and
- adjust traffic light timings dynamically.

Which **architectural style** is best suited for this system? Explain in detail how it works in this context. (4 marks)

ANS:The best architectural style for this scenario is the **Event-Based / Publish-Subscribe (Pub/Sub) architecture**.

How it works in this context:

- **Sensors (publishers):** Devices on roads, vehicles, and traffic lights continuously generate traffic data (e.g., vehicle count, speed, congestion level).
- **Message Broker / Middleware:** A central broker (e.g., MQTT broker, Kafka) receives data from all sensors and organizes it into topics (e.g., “*Congestion Updates*”, “*Accident Alerts*”, “*Traffic Light Status*”).
- **Subscribers (drivers, control centers, traffic lights):**
 - Drivers’ mobile apps subscribe to congestion and accident topics to receive route alerts.
 - Traffic control centers subscribe to system-wide updates to analyze traffic flow.
 - Smart traffic lights subscribe to local congestion topics to adjust signal timings automatically.

Question 3: A bank develops an application where customers use a desktop client to prepare transactions (e.g., transfers, payments). Once completed, the application sends data to the bank's central database for processing. Identify the system architecture used and State one advantage and one disadvantage of this architecture. (4 marks)

ANS: 2-Tier Client-Server Architecture.

Advantage: Simple to implement and fast for small-scale systems.

Disadvantage: Scalability is limited – adding more clients increases server load significantly.

Question 4: A text editor uses threads for spell checking, auto-save, and syntax highlighting. Consider three different threading models:

1. User-level threads (ULTs)
2. Kernel-level threads (KLTs)
3. Two-level threading (combination of ULTs and KLTs)

If the auto-save thread performs a disk write (blocking I/O), explain what happens to the other threads in each model. (6 marks)

ANS:

- **User-Level Threads (ULTs):**

All threads are managed in **user space**. If auto-save performs a **blocking system call**, **all threads stop executing** until the I/O completes. blocking affects the entire process.

- **Kernel-Level Threads (KLTs):**

Each thread is managed by the **kernel**. If auto-save **blocks**, only the **auto-save kernel thread** is blocked. Other threads (spell checking and syntax highlighting) continue executing on other CPU cores. True parallelism; blocking does not affect unrelated threads.

- **Two-Level Threads:**

Combines ULTs and KLTs: multiple **user threads mapped to kernel threads**. If auto-save **blocks**, only the **kernel thread handling that user thread** is blocked. Other user threads mapped to other kernel threads continue running. Flexible scheduling, better CPU utilization, and partial protection against blocking issues.

Question 5: When does a decentralized system become distributed? (1 mark)

ANS: A decentralized system becomes distributed *when links are added between nodes.*

Question 6: Determine the type of scalability involved in the following scenarios. (3 marks)

- a) A cloud-based video streaming service adds servers in multiple continents to reduce latency for international viewers.
Answer: Geographical
- b) A multinational corporation integrates regional payroll systems, each managed by different HR departments, into a single distributed platform.
Answer: Administrative
- c) An online education platform prepares to accommodate millions of new students enrolling globally.
Answer: Size
- d) A logistics company adds additional distribution centers in new regions to handle growing delivery requests while maintaining local management.
Answer: Administrative
- e) A real-time multiplayer gaming platform deploys servers close to players in Asia, Europe, and North America to ensure smooth gameplay.
Answer: Geographical
- f) A retail chain upgrades its inventory system to process higher volumes of orders during holiday seasons by adding more backend servers.
Answer: Size

Question 7: A multinational company wants to integrate its software systems across different locations, departments, and platforms. Each department uses different hardware, operating systems, and programming languages. The company also plans to allow third-party vendors and partners to access some of its resources securely.

- a) Identify whether this is an example of an **open distributed system** or a **closed distributed system**. (1 mark)
- b) Explain the key characteristics of open distributed systems that make this integration possible. (4 marks)

ANS: a) open DS

- c) Systems should conform to well-defined interfaces

Systems should easily interoperate

Systems should support portability of applications

Systems should be easily extensible

Question 8: One of the techniques for scaling is replication. Explain the main problem that replication can cause. (3 marks)

Ans: Having multiple copies (cached or replicated), leads to inconsistencies: modifying one copy makes that copy different from the rest.

Always keeping copies consistent and in a general way requires global synchronization on each modification.

The End